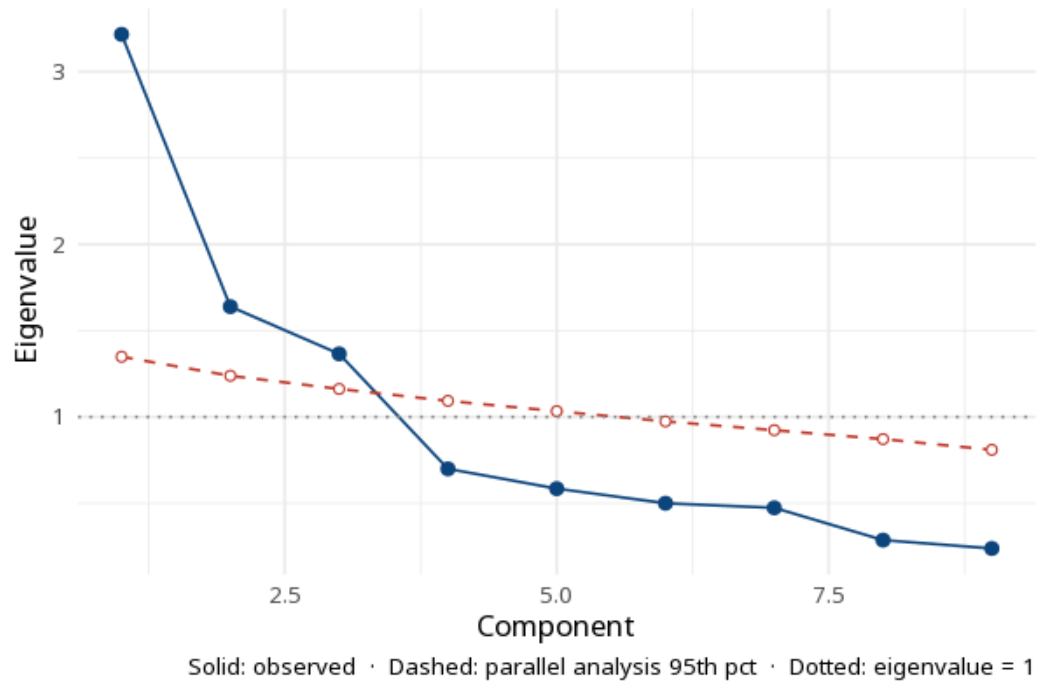


1 Principal component analysis (PCA)

Component	Eigenvalue	% variance	Cumulative %
PC1	3.216	35.7	35.7
PC2	1.639	18.2	53.9
PC3	1.365	15.2	69.1

Variable	PC1	PC2	PC3
x1	.66		−.37
x2	.39		−.62
x3	.48	.33	−.54
x4	.77	−.45	
x5	.74	−.48	
x6	.77	−.43	
x7	.35	.50	.59
x8	.45	.61	.33
x9	.59	.51	



PCA compresses correlated variables into a few components. The variance explained (Table 1) shows how much information each component retains; the component loadings (Table 2) show how strongly each variable contributes. Loadings are correlation-scaled ($\text{eigenvector} \times \sqrt{\text{eigenvalue}}$) so they share the unit of a correlation coefficient. PCA is data reduction — components are

weighted composites of the observed variables, not reflective latent factors; do not interpret them as constructs and do not report communalities (which belong to factor analysis, not PCA). For a latent-construct model use EFA (psych::fa) or CFA (lavaan). The scree plot overlays the parallel-analysis 95th-percentile threshold — retain components where the observed eigenvalue exceeds that line; the Kaiser eigenvalue > 1 rule is shown for reference but tends to over-extract (Zwick & Velicer, 1986). Cumulative variance $\geq 70\%$ is a common but explicitly subjective target (Jolliffe & Cadima, 2016).

PCA (correlation matrix; parallel analysis) retained 3 components explaining 69.1% of total variance.

Statistical basis: Principal component analysis as a data-reduction method (not a latent-variable model) (Jolliffe, I. T., Cadima, J. (2016). "Principal component analysis: a review and recent developments." *Philosophical Transactions of the Royal Society A*, 374, 20150202.); PCA is not EFA - components are composites, not reflective factors (Frick, S., Killisch, R., BiSSant, N., Wetzel, E. (2025). "PCA Is Not EFA." *European Journal of Psychological Assessment*, 41(6), 425-426.); Kaiser eigenvalue > 1 rule tends to over-extract (Zwick, W. R., Velicer, W. F. (1986). "Comparison of five rules for determining the number of components to retain." *Psychological Bulletin*, 99(3), 432-442.).